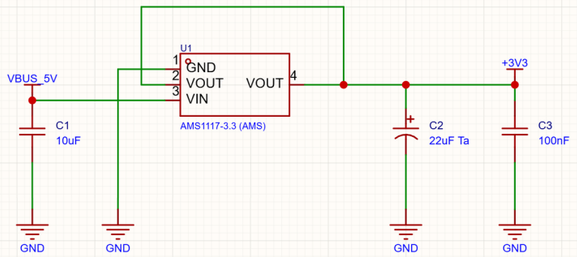
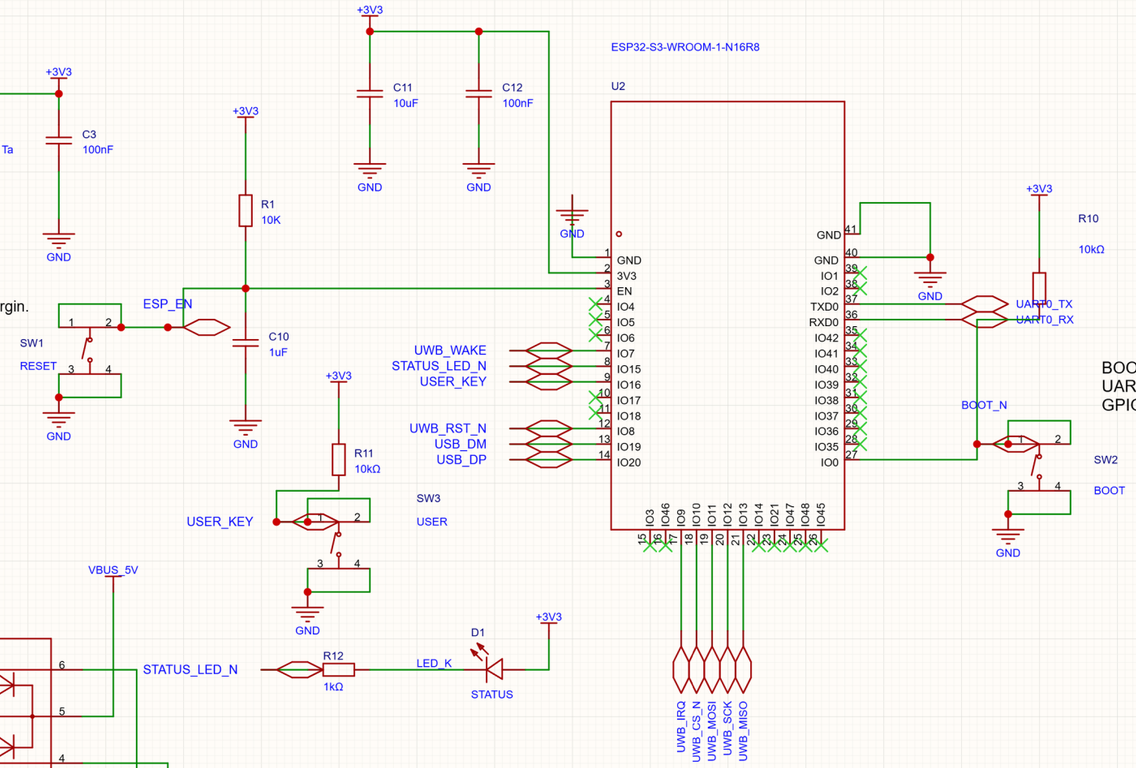
UWB Anchor | ESP32-S3-WROOM-1-N16R8 + DW3220 | DRAFT

USB 供电与 3.3 V 稳压

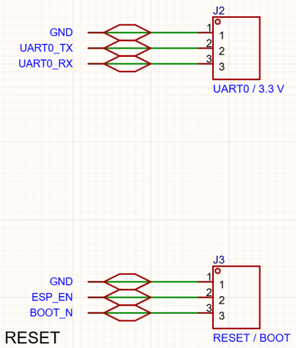


[SPEC] AMS: COUT = 22 uF solid tantalum; tab = VOUT.
[TBD-MEASURE] USB source capability, inrush, dropout and thermal margin.
5 V -> 3.3 V: 0.51 W @ 300 mA; 0.85 W @ 500 mA (plus bias).

ESP32-S3 最小系统 / EN、BOOT 与恢复路径



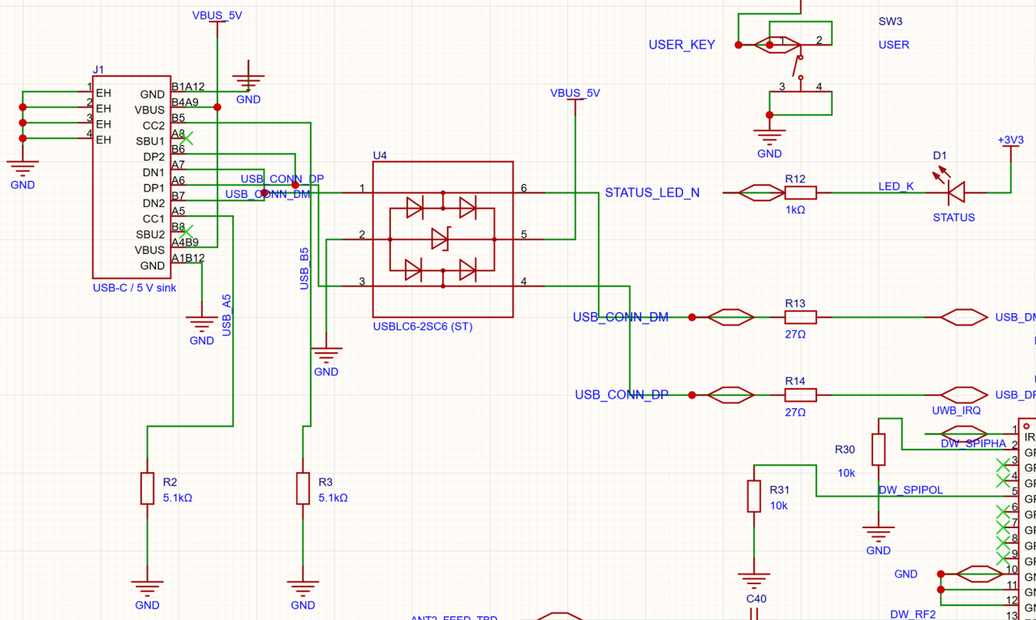
UART0 / 恢复 / 测点



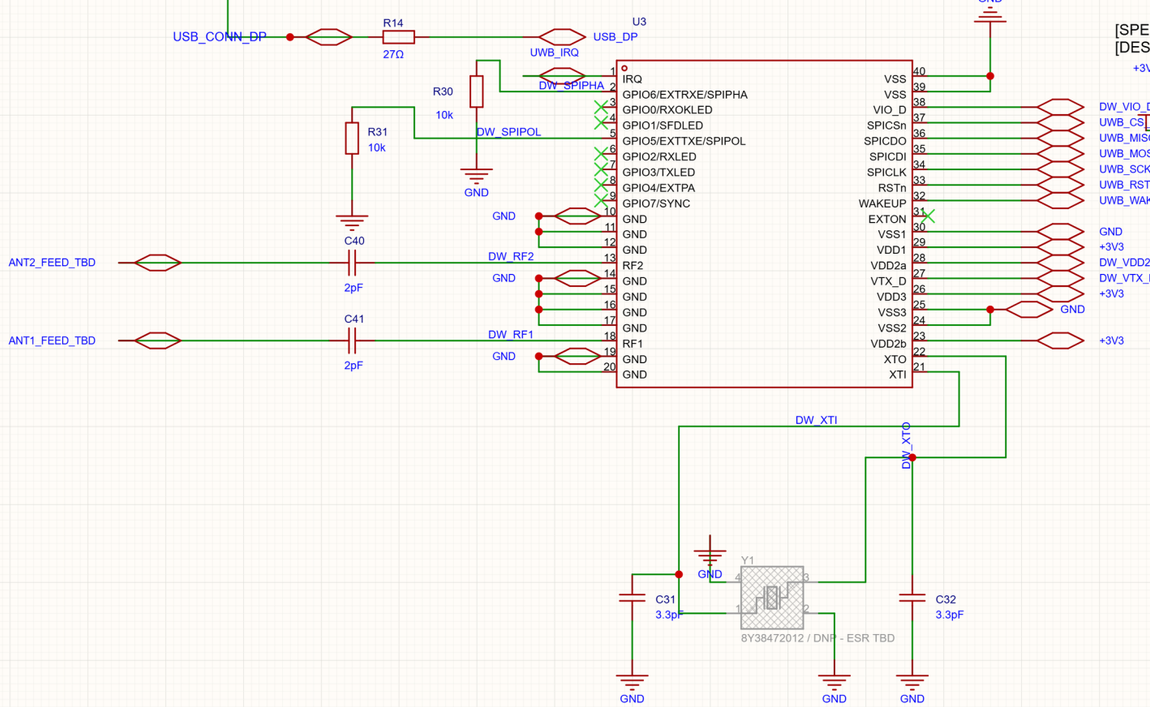
BOOT: hold SW2, pulse RESET
UART0: 3.3 V only
GPIO35/36/37 reserved for PSRAM

No 5 V UART. Connect GND before signal.
USB OTG and Serial/JTAG share the internal PHY.
Do not back-power an unpowered board from UART.

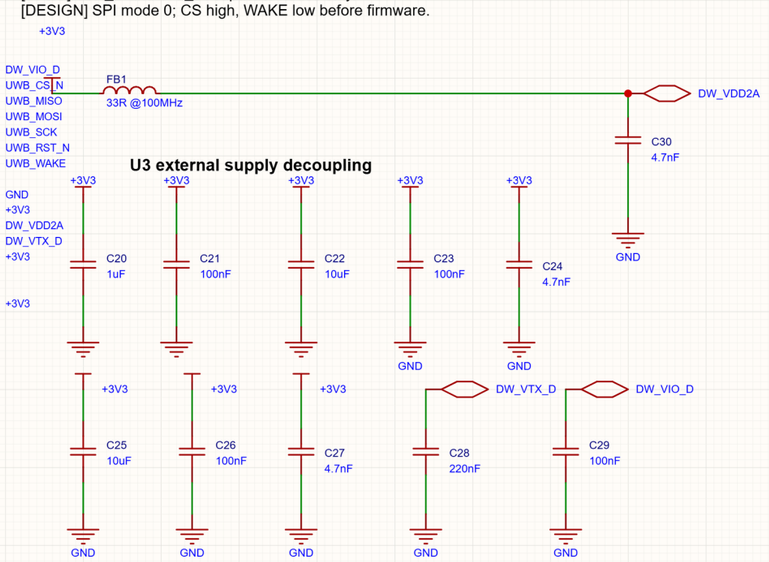
USB-C USB2.0 / ESD 保护

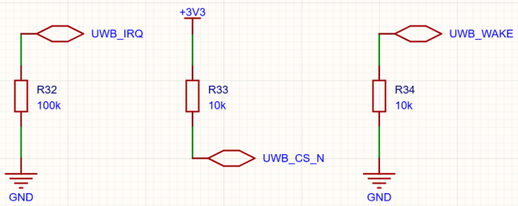


DW3220 / SPI / 38.4 MHz



[SPEC] VTX_D and VIO_D: capacitor nodes only.
[DESIGN] SPI mode 0; CS high, WAKE low before firmware.





双天线馈电 / DRAFT

ANT1 / ANT2 physical design: TBD (user confirmed).
[SPEC] 50 ohm RF1/RF2; 2 pF series DC block.
[TBD-MEASURE] match, mutual coupling and phase centers.
No verified 16.9 mm phase-center spacing; stackup TBD.

RST_N: MCU open-drain LOW / release only.
No external pull-up; no push-pull HIGH.

Y1 候选晶体暂不装配：参考手册推荐该型号，
但当前 ESR 80 ohm 与表 10 的 60 ohm 条件有差异；
需厂家确认或起振/温度/电压裕量验证后定版。